

ARGUS

Autonomous Regional Geospatial Urban Sentinel

A Multimodal Geospatial Intelligence and Predictive Decision-Support System for Closed-Loop Urban Infrastructure Management

Final-Year B.Tech Engineering Project — Detailed Technical Specification v2.0

Pilot Deployment Zone: VIT Bhopal University Campus + Kothri, Madhya Pradesh

Architecture Principle: Zero Recurring Cost · Local-First Inference · Fully Open-Source Stack

Primary Deep Learning Framework: PyTorch | Geospatial Engine: PostGIS | Digital Twin: CesiumJS

Document Guide

This specification expands the original ARGUS blueprint into a full research-grade final-year engineering document. It adds: a positioned literature/novelty case, a concrete zero-cost dataset acquisition plan, a graph-neural-network spatio-temporal risk model as the project's headline research contribution, an explainability layer, a hardened security/privacy design, a week-by-week 20-week execution roadmap, an ablation-driven evaluation protocol, and a fully itemised proof that the entire system can be built and demonstrated for ₹0 in recurring cost. Every subsystem below is written so it can be defended individually in a viva: what it does, why it was chosen over alternatives, how it is evaluated, and what its honest limitations are.

1. Executive Overview

ARGUS is a closed-loop geospatial decision-intelligence platform for urban infrastructure hazards — primarily potholes, road surface degradation, and fallen-tree obstructions. It fuses ground-level citizen photographic evidence, passive civic text signals, satellite/environmental context, weather data, road-network topology, and historical hazard trajectories into a single, explainable operational record per hazard. Perception output feeds severity estimation; severity feeds a transparent multi-factor risk score; risk feeds a predictive forecasting layer; forecasts feed prioritisation and crew dispatch; completed repairs feed an AI-verified closure step; and every closed case re-enters the historical dataset that retrains the predictive layer. Nothing in the loop is a dead end.

The system is deliberately scoped to answer three escalating operational questions, in increasing order of difficulty and value to a municipal supervisor:

- Descriptive — “What is happening right now?” (live hazard map, severity, confidence)
- Predictive — “What is likely to happen next?” (7-day and 30-day spatio-temporal risk forecasts)
- Prescriptive — “What should we do about it, with the resources we actually have?” (priority-aware crew routing, what-if simulation)

Why this scope raises the ceiling on a final-year project

Most student infrastructure projects stop at descriptive detection (“here is a pothole”). ARGUS is designed so the headline contribution — a spatio-temporal Graph Neural Network that propagates risk across the road network graph rather than treating each hazard as an isolated point — is genuinely defensible as research, while every other subsystem remains buildable by a single engineer in one semester using entirely free tooling.

Pilot geography is intentionally constrained to VIT Bhopal and Kothri. This keeps data collection, annotation, validation, and live demonstration tractable while the architecture — service boundaries, database schema, API contracts — is written to scale to a full city without redesign.

2. Problem Statement, Motivation & Research Positioning

2.1 The Reactive Maintenance Problem

Urban road-hazard management in most Indian municipalities, and in most existing citizen-reporting apps, follows a purely reactive lifecycle: a defect is noticed, reported through a disconnected channel (phone call, generic complaint portal, or social media post), manually triaged by an overloaded municipal office, assigned informally to a crew, repaired at an arbitrary priority, and closed without verification. This lifecycle has four structural gaps that ARGUS targets directly:

- Duplication gap — the same pothole is reported dozens of times as unrelated tickets, wasting triage effort.
- Prioritisation gap — severity and urgency are judged subjectively, with no consistent, auditable scoring.
- Foresight gap — maintenance is triggered only after a hazard is already reported, never before it forms from a visible risk pattern (poor drainage, high traffic load, recurring monsoon exposure).
- Verification gap — repairs are marked “done” administratively, with no independent evidence the fix actually resolved the hazard.

2.2 How Existing Approaches Fall Short

Three broad categories of prior work exist, and ARGUS is positioned deliberately against each:

Category	Representative approach	Key limitation ARGUS addresses
Citizen complaint apps	Government grievance portals, generic “report-a-problem” apps	No AI triage, no deduplication, no spatial reasoning, no verification — a form, not an intelligence system
Academic pothole-detection papers	Single-modality CNN/YOLO pothole detectors on curated datasets	Stop at detection; no fusion with context, no forecasting, no operational loop back to a dispatch/verification system
Commercial road-asset platforms	Enterprise GIS + IoT sensor fleets for municipal asset management	Require paid sensor hardware, LiDAR vans, and commercial cloud GIS licences — infeasible for a student budget or a pilot campus

2.3 Research Positioning & Novelty Statement

ARGUS's core academic claim is narrow and defensible, which is precisely what makes it credible: individual hazard severity is not independent of its neighbours on the road graph. A pothole's effective risk is elevated when it sits near other recent defects, on a high-traffic segment, in a drainage-poor zone, ahead of monsoon weather. This is a graph propagation problem, not a per-image classification problem. The project's headline contribution is therefore a spatio-temporal Graph Neural Network (GNN) that models the road network as a graph — nodes are road segments, edges are physical adjacency plus shared-drainage-basin relationships — and propagates a learned risk signal across neighbouring segments over a rolling time window, compared rigorously against the non-graph baselines (moving average, LightGBM/XGBoost, LSTM) specified in the original blueprint.

Defensible novelty claim for the viva

“We do not claim a new pothole detector. We claim that fusing per-hazard detections into a road-network graph and propagating risk with a GNN produces measurably better 7-day forecasts than treating hazards as

spatially independent points — and we prove it with an ablation against three non-graph baselines on the same pilot dataset.” This framing is honest, scoped, and directly evaluable — exactly what a supervisor wants to hear.

3. Objectives

#	Objective	ARGUS capability	Success metric
O1	Multimodal fusion	Combine image, text, GPS, weather, GIS, and environmental evidence into one hazard record	≥ 90% of test hazards carry all applicable evidence fields populated
O2	Hazard characterisation	Detect hazard type and generate a transparent, decomposable severity score	mAP ≥ 0.6 on pilot detection set; severity rank correlation ≥ 0.7 vs human raters
O3	Spatio-temporal prediction	Forecast 7-day and 30-day segment-level risk using the GNN model vs baselines	GNN reduces MAE by a statistically significant margin over the best non-graph baseline
O4	Closed-loop verification	Classify repair state from before/after evidence with confidence	Accuracy ≥ 0.8 on a held-out before/after set
O5	Passive civic mining	Extract hazard/location signals from public text when live feeds are unavailable	NER F1 ≥ 0.75 on an annotated civic-text sample
O6	Dynamic dispatch	Assign and route a limited crew fleet against priority-weighted hazards	≥ 15% total-distance improvement over nearest-neighbour baseline
O7	Immersive visualisation	Expose the full operational picture through a 3D digital twin with temporal playback	Live demo runs end-to-end in under 5 minutes with no manual intervention
O8	Zero recurring cost	Build and demonstrate the entire pipeline without any paid API or managed cloud service	₹0 monthly spend, itemised in Section 15

#	Objective	ARGUS capability	Success metric
Success condition A new hazard report can travel through the complete pipeline — ingestion → geospatial normalisation → AI perception → fusion → GNN risk propagation → map visualisation → crew dispatch → repair verification → audit/history — with a measured, reported evaluation metric captured at every stage, not just at the end.			

4. System Architecture

4.1 Architectural Philosophy

ARGUS is a modular, local-first platform. Heavy model inference runs on the developer machine (or a free-tier cloud GPU session for training only, e.g. Google Colab / Kaggle Notebooks), so the running system never depends on a paid per-request inference provider. Services are split along domain boundaries so any individual model can be replaced, retrained, or upgraded — for instance swapping the detector backbone or the GNN's message-passing layer — without touching the dashboard or the database schema.

4.2 Layered Reference Architecture

- Presentation layer — Next.js + Tailwind CSS + CesiumJS 3D globe for the command centre; a lightweight installable PWA for citizen reporting and field-crew operations (offline-capable via IndexedDB queueing).
- API / contract layer — FastAPI exposing a stable REST + WebSocket contract; this is the only layer the frontend ever talks to, so backend workers can change independently.
- Intelligence layer — five independent Python workers: vision-worker, nlp-worker, forecast-worker (hosting the GNN and baselines), verification-worker, routing-worker.
- Geospatial + persistence layer — PostgreSQL 16 with the PostGIS extension for all spatial reasoning; object storage on local disk (or a free-tier S3-compatible bucket such as Cloudflare R2's free allowance) for media evidence.
- Messaging layer — Redis for job queues between the API and workers, and as a pub/sub backbone for WebSocket event fan-out.
- Deployment layer — Docker Compose orchestrating all services locally for development and live demo; optional single free-tier deployment (Render/Railway free web service + a free Neon/Supabase Postgres-PostGIS instance) for a public demo URL.

4.3 Event-Driven Flow

NEW_REPORT → schema + geofence validation → vision-worker inference → HAZARD_UPDATED → fusion + ARGUS Score → PRIORITY_UPDATED → GNN forecast refresh (batched, not per-event) → optional routing-worker dispatch → REPAIR_SUBMITTED → verification-worker inference → VERIFICATION_COMPLETED →

case closed / reopened → audit_events append (immutable) → nightly retraining snapshot appended to the historical dataset.

4.4 Service Boundaries

Worker	Responsibility	Key libraries
vision-worker	Hazard detection, severity feature extraction, monocular depth proxy	PyTorch, torchvision, OpenCV, MiDaS/Depth Anything
nlp-worker	Civic text NER, classification, urgency/sentiment cues	Hugging Face Transformers
forecast-worker	Feature engineering, baseline models, GNN training + inference	scikit-learn, XGBoost/LightGBM, PyTorch Geometric
verification-worker	Before/after alignment and repair-state classification	OpenCV, a Siamese embedding network in PyTorch
routing-worker	Priority-aware vehicle routing for crew dispatch	OR-Tools, NetworkX

5. AI / ML Methodology

PyTorch is the primary deep-learning framework throughout. The project deliberately mixes deep learning, classical ML, graph learning, and combinatorial optimisation because the underlying problem genuinely spans vision, language, tabular, spatial-graph, temporal, and routing components — a mix that itself demonstrates broader engineering maturity than a single-model project.

5.1 Computer Vision — Hazard Detection & Severity

A PyTorch object-detection model (YOLOv8-nano or a fine-tuned Faster R-CNN, both free and locally trainable on a consumer GPU or free Colab GPU) identifies potholes, road surface cracking, and fallen-tree obstructions in citizen-submitted images. Severity is derived from a feature vector rather than a single opaque score, so it stays explainable:

- Affected area — segmentation-mask pixel area normalised by estimated road-width-in-frame
- Shape irregularity — contour compactness and edge roughness
- Relative depth proxy — from the monocular depth model, never presented as an absolute measurement
- Road occupancy — fraction of drivable lane width affected
- Detector confidence — the model's own calibrated confidence score

Honest constraint (state this explicitly in the report)

Monocular depth is a relative severity cue, not a certified physical measurement, unless a reliable scale reference (e.g. a known-size object, or stereo/LiDAR input) is present in frame. Claiming centimetre-accurate depth from a single RGB photo would be an overclaim a supervisor should flag — so ARGUS never makes that claim, and treats this constraint as a design feature to discuss in the viva.

5.2 Depth Estimation

MiDaS (small variant, CPU-friendly) or Depth Anything is evaluated as a pretrained, frozen depth component — no depth-specific training data is required, which keeps the dataset burden focused on detection and severity labelling.

5.3 NLP / Civic Signal Mining

Hugging Face Transformers (a distilled model such as DistilBERT fine-tuned for a few epochs on a small annotated sample, or a zero-shot pipeline where labelled data is scarce) converts unstructured civic text — scraped public grievance posts, campus complaint-group messages the team has permission to use, or a synthetic/replayed dataset — into structured signals: hazard type, location entity, urgency cue, sentiment/context, and extraction confidence.

5.4 Predictive Modelling — The Headline Research Component

This subsystem is built in four escalating stages, each a legitimate deliverable on its own, so the project always has a working forecaster even if the most advanced stage runs out of semester time:

1. Stage A — Naive baseline: historical rate / moving average per road segment. Establishes the floor every later model must beat.

2. Stage B — Tabular ML: LightGBM or XGBoost on engineered features (recent hazard count, weather exposure, traffic proxy, days since last repair, seasonality). Strong, fast, and highly explainable via SHAP.
3. Stage C — Temporal deep learning: an LSTM or temporal convolutional network, included only if the pilot's time-series depth genuinely justifies it (see Section 12 risk register) — evaluated fairly against Stage B, not assumed superior.
4. Stage D — Spatio-temporal Graph Neural Network (headline contribution): road segments are nodes; edges connect physically adjacent segments and segments sharing a drainage basin or historically correlated hazard patterns. A Graph Convolutional or Graph Attention Network (via PyTorch Geometric, free and open-source) passes messages across this graph over a rolling multi-week window, so a cluster of new potholes on one segment measurably raises the predicted risk of its untouched neighbours before they are ever reported.

Why the GNN stage is the single highest-impact addition

It converts ARGUS from “a system that reacts to reports” into “a system that anticipates hazards before citizens report them,” which is the exact claim municipal infrastructure research cares about — and it is buildable with entirely free, well-documented open-source tooling (PyTorch Geometric) on the same pilot dataset the rest of the project already needs.

All four stages report the same metric set — MAE, RMSE, and where relevant MAPE and calibration/uncertainty — on an identical held-out time split, so the comparison in the final report is a clean, single evaluation table rather than four disconnected experiments.

5.5 Explainable AI Layer

Every model that feeds an operational decision exposes an explanation, not just a number:

- Severity and the ARGUS Score decompose additively into named contributing factors (see Section 6.2's explainability panel).
- The Stage B tabular model exposes SHAP feature-importance values per prediction.
- The GNN exposes attention weights (if a Graph Attention Network is used) showing which neighbouring segments most influenced a given forecast — directly visualisable on the 3D map as “why this hotspot is rated this way.”

5.6 Repair Verification

OpenCV performs perspective alignment and normalisation of before/after images (matching camera angle and lighting as closely as possible from citizen-submitted photos), followed by a comparison combining classical structural similarity (SSIM) with a learned Siamese embedding network. Output is one of Resolved / Partially Resolved / Not Resolved / Uncertain, with a confidence score — never a bare boolean, so field supervisors can see when the AI itself is unsure and route to human review.

5.7 Routing & Dispatch

Dispatch is formulated as a priority-aware capacitated vehicle-routing problem. Three tiers are implemented and compared: a nearest-neighbour baseline, an OR-Tools constraint-programming solver (free, Google-maintained, no licence cost), and — as an optional advanced experiment retained only if it can be evaluated

fairly — a reinforcement-learning dispatch policy. RL is explicitly not a launch requirement; it is a stretch goal that should not put the core deliverable at risk.

6. Multimodal Fusion & the ARGUS Urban Risk Score (AURS)

6.1 Unified Hazard Record

The single most valuable architectural feature is not any individual model — it is the fusion layer that turns heterogeneous evidence into one operational hazard record: hazard ID, type, coordinates, source(s), timestamps, detection confidence, severity, recurrence count, road context, weather exposure, GNN-derived forecast, estimated response effort, repair state, and evidence links. This record is what a municipal supervisor actually reads.

6.2 AURS Formula & Explainability

$AURS = w_1 \cdot \text{Severity} + w_2 \cdot \text{Exposure} + w_3 \cdot \text{Recurrence} + w_4 \cdot \text{WeatherRisk} + w_5 \cdot \text{GNN_Forecast} + w_6 \cdot \text{SourceConfidence}$, with every term normalised to 0–100 and every weight documented in a versioned config file rather than hidden in code — a small detail that itself signals engineering discipline to a supervisor reviewing the codebase.

Example explainability panel shown on the hazard detail view:

Factor	Contribution
Severity	+24
GNN 7-day forecast risk	+21
Historical recurrence	+16
Traffic exposure	+14
Weather risk (upcoming monsoon window)	+11
Multiple corroborating reports	+9
Source confidence	+5
TOTAL	= 100 / 100

6.3 Duplicate & Corroboration Intelligence

Twenty-seven citizen reports of the same pothole must not become twenty-seven independent tickets. Spatial clustering (PostGIS ST_ClusterDBSCAN), temporal proximity windows, and image-embedding similarity collapse corroborating observations into one hazard with multiple evidence sources; the number and reliability of corroborating observations then strengthen — never solely determine — the hazard's confidence.

6.4 Citizen Trust Layer

A reporter reliability score, built from verified-report rate, false-positive rate, duplicate rate, location accuracy, and historical outcome agreement, adjusts confidence gradually. It is designed to never let a single low-trust reporter suppress or dominate a hazard's score, which matters both for robustness and for the ethical/security discussion in Section 10.

7. Geospatial Intelligence

PostGIS is the central spatial reasoning engine, not merely a place to store latitude/longitude pairs. Core operations: spatial de-duplication, nearest-road-segment matching, hazard-to-segment linking, radius queries, DBSCAN hotspot clustering, temporal aggregation windows, and map-ready geometry generation (GeoJSON) served directly to the Cesium frontend.

7.1 Multi-Resolution Evidence Strategy

- Ground imagery — fine-grained, per-hazard evidence (the primary detector input).
- Sentinel-2 (via the free Copernicus Open Access Hub / Sentinel Hub free tier) — regional/environmental context only, e.g. vegetation and surface-water proxies, never a direct per-pothole detector.
- OpenStreetMap (free, no API key required for the standard tile/data usage this pilot needs) — road-network topology and segment metadata.
- OpenWeatherMap free tier — current and historical weather features feeding both the AURS WeatherRisk term and the forecast-worker's temporal features.
- SRTM / open elevation data — terrain context for the Cesium 3D visualisation and drainage-basin edge construction in the GNN's road graph.

7.2 Hotspot Example

Cluster #17: 13 corroborating observations within one spatial corridor, active for three weeks, high severity, elevated upcoming weather exposure, and a GNN forecast flagging two adjacent untouched segments. ARGUS converts this from thirteen isolated complaints into one recurring, forecast-augmented infrastructure-risk hotspot — precisely the kind of insight a manual complaint log cannot produce.

7.3 Temporal Playback

The command-centre dashboard includes a date slider replaying hazards as they appear, worsen, get repaired, and (occasionally) reappear — making the predictive and closed-loop claims visually demonstrable in under a minute rather than requiring the supervisor to trust a static metrics table.

8. Zero-Cost / Local-First Technology Strategy

Every layer below has a free or local substitute with no recurring cost, and the pipeline is designed to keep functioning if any single free tier is temporarily unavailable — a resilience property worth stating explicitly in the report.

Layer	Free / local choice	Free-tier limit to know	Fallback if exceeded
Deep learning	PyTorch, trained locally or on free Colab/Kaggle GPU sessions	Colab free GPU sessions time out after ~12h idle/continuous use	Checkpoint + resume training across sessions
Vision utility	OpenCV, Albumentations (augmentation)	None — fully local	—
NLP	Hugging Face Transformers, distilled models	None for inference; fine-tuning needs a GPU session	Use CPU inference for small distilled models
Graph learning	PyTorch Geometric	None — fully local	—
Prediction baselines	scikit-learn, XGBoost/LightGBM	None — fully local	—
Database	PostgreSQL + PostGIS, self-hosted; free-tier Neon/Supabase for a public demo	Supabase/Neon free tier: ~0.5–1GB storage	Prune old media blobs; keep only metadata + thumbnails in the free tier
Object storage	Local disk; Cloudflare R2 free allowance for demo	R2 free tier: 10GB storage / 1M reads / mo	Keep pilot-scale evidence well under this ceiling
Cache / jobs	Redis, self-hosted via Docker	None — fully local	—
Maps	OpenStreetMap	Fair-use tile request limits on public tile servers	Self-host tiles or cache aggressively for the demo
Satellite / GIS	Sentinel-2 via Copernicus/Sentinel Hub free tier	Free tier request quota per month	Cache regional context images once; refresh rarely
Weather	OpenWeatherMap free tier	60 calls/min, 1,000 calls/day	Batch and cache weather pulls per segment per day
3D visualisation	CesiumJS (open-source Cesium ion)	Cesium ion free tier has a	Use offline/local terrain tiles for

Layer	Free / local choice	Free-tier limit to know	Fallback if exceeded
	free asset tier for terrain/imagery)	monthly streaming quota	the live demo to avoid quota risk
Routing optimisation	Google OR-Tools	None — fully local, Apache-2.0 licensed	—
Hosting (optional public demo)	Render/Railway free web service tier	Free tier apps sleep after inactivity	Wake-on-request is acceptable for a scheduled demo, not for 24/7 uptime
CI/CD	GitHub Actions free minutes for a student/public repo	2,000 free minutes/month on private repos (unlimited on public)	Keep the repo public where feasible
Packaging	Docker + Docker Compose	None — fully local	—
Development assistance	Any available free coding assistant	Varies by tool	Fall back to manual development

Cost-control rule

APIs are used only where they add clear value and stay within a documented free tier. Core perception, scoring, forecasting, and routing never depend on a paid per-request provider — the system's intelligence lives in code the team owns and can run offline.

9. Dataset Creation & Annotation Plan (Zero Cost)

A credible final-year project is judged as much on data discipline as on model choice. ARGUS's dataset plan is built entirely from free sources plus the team's own free data collection effort:

- Public seed datasets — freely available road-damage datasets (e.g. RDD2022-style regional road-damage collections, open Kaggle pothole datasets) bootstrap the detector before any campus-specific images exist.
- Campus-specific collection — structured phone-camera walkthroughs of VIT Bhopal and Kothri at scheduled intervals (e.g. weekly, and immediately after rain), each image geotagged automatically by the phone's GPS. This is the zero-cost equivalent of a survey vehicle.
- Annotation — CVAT (Computer Vision Annotation Tool), free and self-hostable via Docker, used for bounding boxes/segmentation masks, severity labels, and before/after pairing.

8. Augmentation — Albumentations (free, open-source) synthetically expands the small campus dataset with blur, low-light, rain, and compression variants, directly supporting the robustness evaluation in Section 11.
9. Civic text — a small hand-annotated sample of public grievance-style text, or a synthetic/replayed dataset when live social feeds are unavailable, for NER/classification fine-tuning.

Data quality principle

A smaller, carefully annotated pilot dataset with honest severity labels is worth more — and is far more defensible in a viva — than a large, noisy, unverified scrape. State the exact number of annotated images, the annotation protocol, and inter-annotator agreement (even informally, between two team members) in the final report.

10. Security, Privacy & Ethical Design

A system that ingests citizen-submitted geotagged photos of public infrastructure carries real privacy obligations, and addressing them explicitly is exactly the kind of maturity that distinguishes a strong final-year project from a purely technical demo.

10.1 Data Minimisation & Anonymisation

- Faces and licence plates incidentally captured in road-hazard photos are automatically blurred using a lightweight local detector before storage — never sent anywhere unblurred.
- Precise reporter identity is decoupled from public-facing hazard records; only an internal reliability score is retained, never displayed publicly against a name.
- Raw GPS is snapped to the nearest road segment for public display; exact submission coordinates are retained only in the internal audit trail.

10.2 Access Control

- Role-based access: citizen (submit + view own reports), field crew (assigned-route + repair submission), municipal operator (full command-centre + dispatch), and read-only analytics for stakeholders.
- All state-changing API calls are authenticated and written to the immutable `audit_events` table.

10.3 Model & System Robustness

- Every AI output carries a confidence score, and the UI visually distinguishes an AI prediction from a human-confirmed field observation — never presenting a model guess as ground truth.
- Human override is always available at the dispatch and verification stages; AI recommends, a municipal operator can always overrule.

10.4 Responsible Scope

ARGUS never claims certified physical measurements (e.g. exact pothole depth) from a single RGB image, never fully automates repair crew dispatch without a human-confirmable override, and treats every predictive output as a forecast with reported uncertainty rather than a guarantee.

11. Evaluation & Research Plan

Subsystem	Baseline	Primary metrics	Ablation / comparison
Hazard detection	Pretrained off-the-shelf detector, unmodified	Precision, Recall, F1, mAP	Fine-tuned vs. pretrained-only
Severity scoring	Rule-based score from area/shape only	MAE, RMSE, rank correlation vs. human raters	With vs. without depth-proxy feature
NER / location extraction	Keyword-matching baseline	Precision, Recall, F1, location accuracy	Zero-shot vs. fine-tuned

Subsystem	Baseline	Primary metrics	Ablation / comparison
Forecasting (Stage A–D)	Moving average / historical rate	MAE, RMSE, MAPE, calibration/uncertainty	Naive vs. LightGBM vs. LSTM vs. GNN — the headline comparison
Repair verification	SSIM / classical image difference only	Accuracy, Precision, Recall, F1	Classical-only vs. classical + Siamese embedding
Routing	Nearest-neighbour	Total distance, response time, priority-weighted coverage	Nearest-neighbour vs. OR-Tools vs. (optional) RL
System performance	Synchronous single-request baseline	API latency, inference time, throughput	Sequential vs. queued worker pipeline
Robustness	Clean, well-lit images only	Performance under blur, low light, rain, compression	Clean vs. augmented test set (Albumentations)

The single most important table in the final report is the forecasting comparison (Stage A vs. B vs. C vs. D) on an identical held-out time split — this is the evidence for the project's headline novelty claim, and it should be reported with confidence intervals or a simple significance test (e.g. a paired comparison across the same segments/time windows), not a single point estimate.

12. Implementation Roadmap — 20-Week Execution Plan

The project is built in layers so a working, demonstrable system exists at every checkpoint — never a pile of disconnected notebooks that only come together in the final week.

Weeks	Phase	Build	Exit criterion
1–2	Foundation setup	Repo scaffolding, Docker Compose, FastAPI skeleton, PostGIS schema, Next.js + Cesium shell	An empty map renders and the API health-check responds
3–4	Ingestion & PWA	Citizen report PWA (offline-queue capable), NEW_REPORT endpoint, media storage	A submitted report appears as a raw point on the map
5–7	Perception I	Detector fine-tuning on seed + early campus data, severity feature pipeline	A report shows AI-detected type, severity, and confidence
8–9	Fusion & dedup	Spatial clustering, corroboration logic, AURS v1 with explainability panel	Duplicate reports collapse into one ranked hazard
10–12	Prediction I–II	Weather/temporal features, Stage A + Stage B	A 7-day forecast with reported MAE/RMSE

Weeks	Phase	Build	Exit criterion
		(LightGBM/XGBoost) forecasting	exists per segment
13–15	Prediction III — GNN (headline)	Road-graph construction, PyTorch Geometric GCN/GAT, Stage C/D comparison	GNN forecast beats baselines with a reported, significant margin
16	Prescriptive layer	OR-Tools routing, crew assignment, priority-weighted coverage	3 crews routed against 20+ hazards with measured route quality
17	Closed loop	Repair workflow, before/after verification worker, audit trail	A repair can be verified or reopened end-to-end
18	Command centre polish	Temporal slider, explainability panels, what-if simulator	Full 3D command centre matches the deliverables in Section 13
19	Evaluation & robustness	Run the full metrics table (Section 11), augmentation robustness tests	Every subsystem has a reported baseline-vs-final metric
20	Documentation & demo rehearsal	Final report, viva slides, rehearsed 5-minute end-to-end demo script	Demo runs live, twice, with no manual intervention

Scope discipline

Kubernetes, full city-scale deployment, exact physical cost estimation, fully autonomous field robotics, and 24/7 production uptime are explicitly future-scale concepts, not requirements for the final-year prototype. A dockerized modular system, reproducible experiments, and a convincing five-minute pilot demonstration are the priority — protecting the semester timeline from scope creep.

13. Deliverables & Live Demonstration

13.1 Citizen View

Capture or upload a geotagged hazard photo; see local AI preprocessing feedback immediately; submit; track status through to eventual verified resolution.

13.2 Authority Command Centre

3D Cesium map of live hazards, a risk heatmap layered with GNN-forecast zones, crew locations and routes, and an individual hazard intelligence panel showing the full explainability breakdown from Section 6.2.

13.3 Field Crew View

Assigned route, priority-ordered task list, navigation context, repair-status update, and after-image submission — usable offline in low-connectivity field conditions via the PWA's queued-sync design.

13.4 Verification View

Side-by-side before/after comparison with repair state, confidence, and the explainable evidence behind the AI's classification.

13.5 Executive Analytics

Infrastructure Health Index, active hazard count, resolution rate, average response time, route efficiency, forecast performance, and recurring hotspots — a single dashboard screen a non-technical supervisor can read in thirty seconds.

13.6 What-If Simulator

Adjust crew count or weather assumptions and observe the resulting change in response time, coverage, and GNN-predicted risk — turning the platform from a passive dashboard into a decision-support tool during the live viva.

14. Data Model & API Blueprint

14.1 Core Tables

Table	Purpose
hazards	Canonical hazard entity and geometry — the fused, de-duplicated record
observations	Individual citizen/social raw observations, pre-fusion
media	Image evidence, linked to observations and repairs
hazard_scores	Severity / confidence / AURS outputs, versioned per recalculation
weather_observations	Historical and forecast weather context per segment
road_segments	OSM-derived road graph nodes/edges, used directly by the GNN

Table	Purpose
predictions	Time-indexed forecasts from every model stage (A–D), for direct comparison
crews	Crew availability, current location, capacity
dispatch_runs	Routing optimisation outputs and the metrics used to score them
repairs	Repair action records, linked to before/after media
verifications	Before/after AI verification results and confidence
users / reputation_events	Accounts and the citizen trust layer (Section 6.4)
audit_events	Immutable lifecycle trail for every state-changing action

14.2 Representative API Surface

- POST /reports — submit a new citizen observation
- GET /hazards, GET /hazards/{id}, GET /hazards/nearby?lat=&lon=&radius=
- GET /hotspots — DBSCAN-clustered active hotspots
- GET /risk/forecast?segment_id=&horizon= — returns the Stage A–D forecast comparison for a segment
- GET /crews, POST /dispatch/optimize
- POST /repairs/{id}/verify
- GET /analytics — executive dashboard aggregate metrics

14.3 Real-Time Events

WebSockets publish hazard-created, priority-updated, crew-moved, repair-submitted, and verification-completed events so the 3D command centre updates live during the demo without a manual page refresh — a small detail that reads as production-grade polish to a supervisor watching the demo.

15. Itemised Zero-Cost Proof

To make the “free to build” claim auditable rather than a slogan, every component's cost is stated explicitly:

Component	Monthly cost	Justification
Development machine compute	₹0	Team's own laptop/desktop; no cloud training required for pilot scale
Model training (optional GPU boost)	₹0	Free Google Colab / Kaggle Notebooks GPU sessions
Database (dev)	₹0	Self-hosted PostgreSQL + PostGIS via Docker
Database (public demo, optional)	₹0	Neon or Supabase free tier, within documented storage limits
Object storage	₹0	Local disk for dev; Cloudflare R2 free allowance for a public demo
Maps & satellite context	₹0	OpenStreetMap; Copernicus/Sentinel Hub free tier
Weather data	₹0	OpenWeatherMap free tier, cached and batched
3D terrain/imagery	₹0	Cesium ion free asset tier, or offline local tiles for the demo
Routing optimisation	₹0	Google OR-Tools, Apache-2.0 licensed
Hosting (optional public demo)	₹0	Render/Railway free web-service tier
CI/CD	₹0	GitHub Actions free minutes on a public repository
TOTAL RECURRING COST	₹0	Every dependency above has a documented free tier or is fully self-hosted

16. Technical Risks & Mitigations

Risk	Why it matters	Mitigation
Insufficient labelled images	Models may overfit or fail to generalise	Build a focused pilot dataset (Section 9); augment aggressively; evaluate robustness explicitly
Monocular depth scale ambiguity	Exact pothole depth cannot be guaranteed from one RGB image	Use relative depth only as a severity feature; label it as an estimate everywhere in the UI and report
Sparse time-series history	LSTM and GNN may be under-justified with too little historical data	Use strong classical baselines regardless; only report the GNN result alongside honest data-volume caveats
Road-graph construction noise	OSM segment topology may not perfectly match on-ground drainage/traffic reality	Manually verify the pilot-area graph by hand — small enough scope to do so — before trusting GNN edges
Live social-media access	API access, policy, or availability may change	Treat passive civic mining as a modular, swappable component; support replayed/open/synthetic data
RL instability (if attempted)	RL may not outperform deterministic routing	Keep RL strictly optional/experimental; OR-Tools remains the shipped default

Risk	Why it matters	Mitigation
Free-tier quota exhaustion	A demo could hit a rate limit at the worst moment	Cache aggressively, batch external calls, and rehearse the demo against cached data
False confidence in AI outputs	Incorrect operational recommendations could be trusted blindly	Always show confidence/evidence; keep human override available at every decision point

17. Final Project Vision

ARGUS is not a complaint tracker, and it is not a single detection model wrapped in a dashboard. It is a closed-loop infrastructure intelligence system that observes a city from multiple evidence channels, converts those observations into spatially grounded hazard records, scores risk transparently, propagates that risk across the road network with a graph neural network to anticipate emerging hotspots before they are widely reported, recommends interventions within real resource constraints, verifies whether those interventions actually worked, and feeds every outcome back into the model that made the next prediction better.

The final demonstration makes this loop obvious in under five minutes: report → detect → score → forecast → dispatch → repair → verify → learn — and every step in that sentence corresponds to a working, evaluated piece of the system, not a slide.

Recommended Project Title

“ARGUS: A Multimodal Geospatial Intelligence and Graph-Neural Predictive Decision-Support System for Closed-Loop Urban Infrastructure Management.” The wording accurately reflects both the operational decision-support role and the specific graph-learning research contribution, while leaving room to discuss RL routing and edge inference as acknowledged future work rather than overclaimed present capability.

Closing note for the supervisor meeting

Lead with the loop, not the tech stack: show one hazard travelling all the way from a citizen's photo to a verified repair in the live demo, then open the explainability panel and the GNN forecast comparison table. The architecture and the free-tier discipline are what prove it is buildable in one semester; the GNN ablation table is what proves it is more than a semester project.

END OF PROJECT SPECIFICATION — v2.0

Prepared as the working technical blueprint for ARGUS.